

computed by taking 16-bit words from header fields. The checksum field ensures checking of errors only at link level on header fields.

TCP performs end-to-end error control, which takes care of errors on the data field. That is why at IP level, error checking at the data field is not done. IP header checksum is computed at each router or node, or on hop-to-hop basis, since each router/node changes the TTL field as well during fragmentation of IP headers (like identification, M flag, offset fields, etc).

The next two levels are for addresses. IP header has two address fields: source IP address and destination IP address. These are used to route datagrams. Similarly, addresses are used in postal letters. IP uses four modes of addressing for this purpose. These are known as A-class, B-class, C-class and D-class. The difference in classes is due to the number of bits used for network identification and the number of bits used for host identification.

Different address formats under different classes are illustrated in Figure 5. IP addresses are of 4 bytes, and these are usually mentioned as four sets of dot-separated octets. For A-class, most significant bit of the first octet is always 0. Therefore the next seven bits are used for network identification. This means that only 27 to 128 networks can be of A-class. Hosts may be 224.

For B-class address, first two bits of the first octet must be 10. This means the first octet has to be from 128 to 191. 127 is reserved for special purposes. Total networks and hosts under this class are, respectively, 214 and 216.

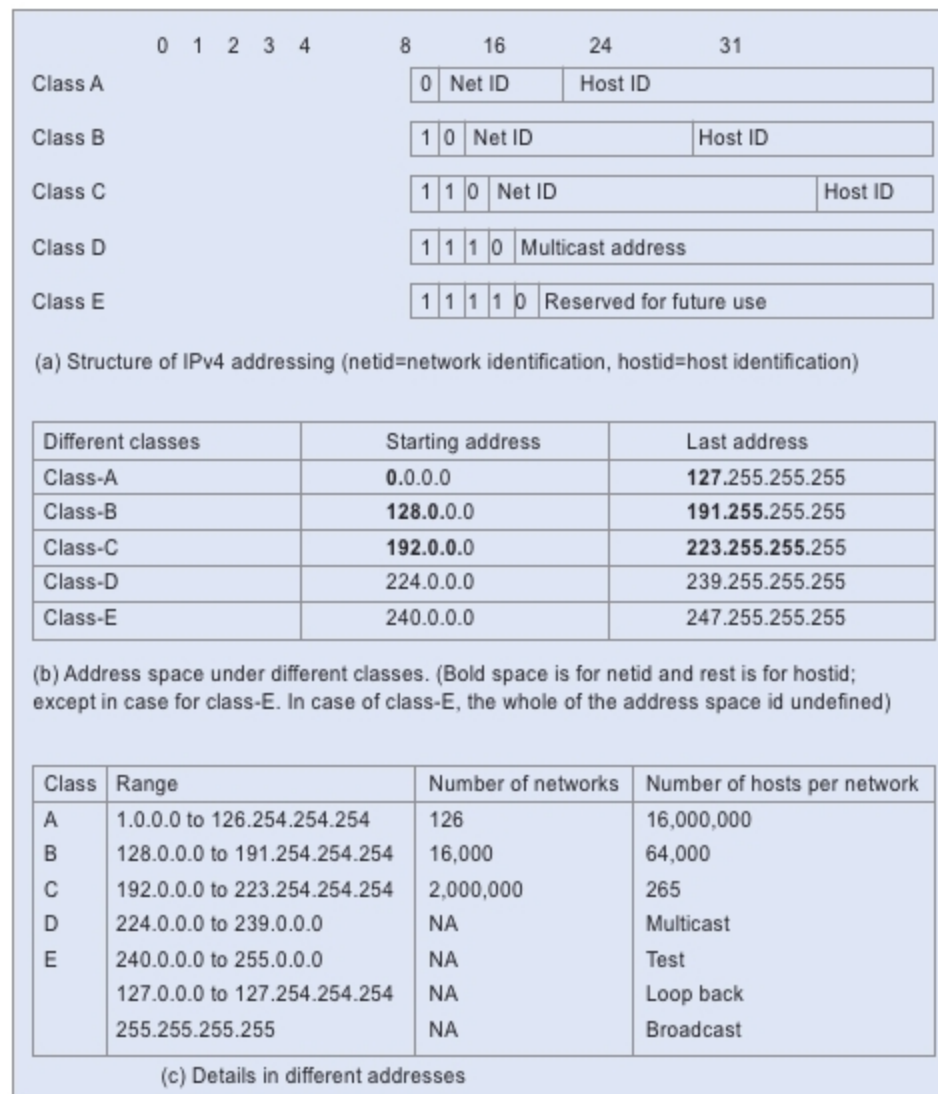


Figure 5: Illustration of IPv4 address scheme

For C-class address, first three bits of the first octet are 110. This means the first octet will be from 192 to 255. Class-C address has 221 networks and 28 hosts. Minimum hosts under different addressing modes are:

$$\text{Class A: } 2^7 \times 2^{24} = 2^{31}$$

$$\text{Class B: } 2^{14} \times 2^{16} = 2^{30}$$

$$\text{Class C: } 2^{21} \times 2^8 = 2^{29}$$

Address scheme under A-class to C-class is for unicast (one-to-one) communication. Other addressing schemes are for multicast (one-to-many) communication. Class-D and Class-E belong to multicasting.

In D-class, all four octets are used to identify the group of nodes designated to receive a multicast. D-class addresses do not specify the network. These are in the range of 224.0.0.0 to 239.255.255.

E-class addresses range from 240 to 255 in the first octet, and are used for experimentation.

Maximum number of hosts or nodes per network are in A-class, and is about 2^8 times that of B-class and about 2^{16} times that of C-class. Maximum number of hosts or nodes per network in B-class

is about 2^8 times that of C-class. The logical conclusion is that A-class addressing is for larger networks, B-class for medium and C-class for small networks. So, if your network is large, you can apply for A-class address, for medium network, B-class and for small network, C-class address.

An organisation may opt for any one of these classes based on size. Say, an organisation has 2000 hosts under its local network. Thus, it cannot opt for C-class address, because under one network address of C-class, a maximum of 128 hosts can be addressed. The optimal option is then for B-class, because under a network address of B-class, maximum of 2^{16} hosts can be addressed.

Special addressing schemes are as follows:

- Address 0.0.0.0 means that this host is on the network. It actually refers to the Internet itself.
- 255.255.255.255 is used as a broadcast (one-to-all communication) packet for all networks. A network ID with host IDs with all 1s or 255 is used for directed broadcast to the network. For example, 20.255.255.255 is used for a broadcast message to A-class network with ID 20. Broadcast addresses are destination addresses.
- Packets with first octet as 127 are used for network testing. An address with network ID as 127 and host ID of any address is used for internal host loop back address. Thus, loop-back addresses are 127.0.0.0 to 127.254.254.254
- An entire network is specified by providing only the network identification and with 0s in all other octets, such as 125.0.0.0 for A-class network, 129.150.0.0 for B-class network and 208.127.110 for C-class network.